

Direct products & Finitely Generated Abelian Groups

Section 11

Def: The Cartesian product of sets $S_1, \dots, S_n$ is

$$S_1 \times \dots \times S_n = \{(a_1, \dots, a_n) \mid a_i \in S_i, \forall 1 \leq i \leq n\} = \prod_{i=1}^n S_i$$

Thm: Let $G_1, \dots, G_n$ be groups. Then $G_1 \times \dots \times G_n$ is a group under $(a_1, \dots, a_n) * (b_1, \dots, b_n) = (a_1 b_1, a_2 b_2, \dots, a_n b_n)$.

applying
operation
from G_i

Pf:

Identity: let e_i be the identity of G_i , for all i . Then

$(e_1, e_2, \dots, e_n)$ is an identity for $G_1 \times \dots \times G_n$:

$$(e_1, \dots, e_n) * (a_1, \dots, a_n) = (e_1 a_1, \dots, e_n a_n) = (a_1, \dots, a_n).$$

Inverse:

$$(a_1, \dots, a_n)^{-1} = (a_1^{-1}, \dots, a_n^{-1})$$

Associativity:

follows from associativity of the G_i .



Examples:

- V_4 Klein four group

$$\simeq \mathbb{Z}_2 \times \mathbb{Z}_2.$$

$$(0,0) \longleftrightarrow e$$

$$(0,1) \longleftrightarrow a$$

$$(1,0) \longleftrightarrow b$$

$$(1,1) \longleftrightarrow c$$

$$(1,0) + (0,1) = (1,1) \checkmark$$

$$b \cdot a = c$$

$$(1,0) + (1,0) = (2,0) = (0,0)$$

$$b \cdot b = e$$

V_4	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

- $\mathbb{Z}_2 \times \mathbb{Z}_3 \simeq \mathbb{Z}_6.$

is cyclic, generated by $a = (1,1)$.

$$\langle a \rangle = \{ \underset{e}{(0,0)}, \underset{a}{(1,1)}, \underset{a^2a}{(0,2)}, \underset{a^3a^2a}{(1,0)}, \underset{4 \cdot a}{(0,1)}, \underset{5 \cdot a}{(1,2)} \} = \mathbb{Z}_2 \times \mathbb{Z}_3$$

Examples:

- $\mathbb{Z}_n \times \mathbb{Z}_n$ can't be cyclic because for any $(a,b) \in \mathbb{Z}_n \times \mathbb{Z}_n$, (a,b) has order at most n : $(na, nb) = (0, 0)$.

- $\underbrace{\mathbb{Z} \times \mathbb{Z} \times \dots \times \mathbb{Z}}_d = \mathbb{Z}^d$ integer vectors under addition.

Thm: let $(a_1, \dots, a_n) \in \prod_{i=1}^n G_i$. If each a_i has order r_i in G_i , then $(a_1, \dots, a_n)$ has order $\text{lcm}(r_1, \dots, r_n)$.

Pf: In order to have $(a_1, \dots, a_n)^r = (e_1, \dots, e_n)$
" $(a_1^r, a_2^r, \dots, a_n^r)$

we need $a_i^r = e_i$ for all i . So r is a multiple of r_i for all i , and any such multiple works. Hence the order is $\text{lcm}(r_1, \dots, r_n)$. $\square$

Thm: $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic (and therefore isomorphic to $\mathbb{Z}_{mn}$)
iff m and n are relatively prime.

Pf: Fact: $mn = \text{lcm}(m, n) \cdot \text{gcd}(m, n)$ ← Exercise: write out the prime factorizations

§ m, n are relatively prime, so $\text{gcd}(m, n) = 1$. Then

$$mn = \text{lcm}(m, n) = |\langle (1, 1) \rangle|$$

$$\text{So } \langle (1, 1) \rangle = \mathbb{Z}_m \times \mathbb{Z}_n.$$

Conversely, § $\langle (a, b) \rangle = \mathbb{Z}_m \times \mathbb{Z}_n$. Then the order of $(a, b) = mn$, so we have $mn = \text{lcm}(\text{order of } a \text{ in } \mathbb{Z}_m, \text{order of } b \text{ in } \mathbb{Z}_n)$.
so $\text{gcd}(m, n) = 1$. □

Cor: The group $\prod_{i=1}^n \mathbb{Z}_{m_i}$ is cyclic iff $\gcd(m_i, m_j) = 1, \forall i \neq j$.

Pf: Repeated application of previous result.

Examples:

• Find order of $(8, 4, 10)$ in $\mathbb{Z}_{12} \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}$.

order of 8 in $\mathbb{Z}_{12}$: $12/\gcd(8, 12) = 12/4 = 3$

order of 4 in $\mathbb{Z}_{60}$: $60/\gcd(4, 60) = 60/4 = 15$

order of 10 in $\mathbb{Z}_{24}$: $24/\gcd(10, 24) = 24/2 = 12$

• Find order of $((1423), 2)$ in $S_4 \times \mathbb{Z}_3$.

order of $(1423) = 4$

order of 2 in $\mathbb{Z}_3 = 3$

$$\text{lcm}(3, 15, 12)$$

" " "

3 3·5 2²·3

$$2^2 \cdot 3 \cdot 5 = \boxed{60}$$

$$\text{lcm}(3, 4) = \boxed{12}$$

Claim: An n -cycle $(a_1 a_2 \dots a_n)$ has order n .

Generators of direct product groups

Let $\bar{G}_i \leq \prod_{j=1}^n G_j$ be

$$\bar{G}_i = \{ (e_1, e_2, \dots, e_{i-1}, a_i, e_{i+1}, \dots, e_n) \mid a_i \in G_i \}.$$

Then $\bar{G}_i \leq \prod_{j=1}^n G_j$ and $\bar{G}_i \cong G_i$.

Pf: Isomorphism $f_i: G_i \rightarrow \bar{G}_i$ sends $a_i \mapsto (e_1, \dots, e_{i-1}, a_i, e_{i+1}, \dots, e_n)$

Cor: If $G_1, \dots, G_n$ are finitely generated, then so is $\prod_{i=1}^n G_i$.

Pf: Let $\{a_{i,1}, \dots, a_{i,k_i}\}$ be a generating set for G_i . Then

$\{f_i(a_{i,j}) \mid 1 \leq j \leq k_i, 1 \leq i \leq n\}$ is a generating set for $\prod_{j=1}^n G_j$. $\square$

Note:

$$(a_1, \dots, a_n) = (a_1, \overset{\downarrow \overline{G}_1}{e_2}, \dots, e_n) * (e_1, a_2, \overset{\downarrow \overline{G}_2}{e_3}, \dots, e_n) * \\ \dots * (e_1, \dots, e_{n-1}, a_n) \\ \uparrow \overline{G}_n$$

Finitely Generated Abelian Groups

Fundamental Theorem of Finitely Generated Abelian Groups

Thm (FTFAG): Every finitely generated abelian group is isomorphic to a direct product

$$\mathbb{Z}_{p_1^{r_1}} \times \cdots \times \mathbb{Z}_{p_n^{r_n}} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$$

where p_i are primes (not necessarily distinct) and $r_i \in \mathbb{Z}^{>0}$.
This decomposition is unique up to permuting factors.

$$\mathbb{Z}_2 \times \mathbb{Z}_2$$

$$\neq$$

$$\mathbb{Z}_2^2 = \mathbb{Z}_4$$

Example: Find all abelian groups, up to isomorphism,
of order 360.

$$2^3 \cdot 3^2 \cdot 5^1 = 360$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

$$\mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$$

Abelian

Groups of order $16 = 2^4$

$$\mathbb{Z}_{16}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_8$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_4$$

$$\mathbb{Z}_4 \times \mathbb{Z}_4$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$$

Thm: If m divides the order of a finite abelian group G , then G has a subgroup of order m .

Pf: write $|G| = p_1^{r_1} \cdots p_n^{r_n}$ then

$$m = p_1^{s_1} \cdots p_n^{s_n} \quad \text{with } 0 \leq s_i \leq r_i, \forall i.$$

Use FTFTAG and the fact that cyclic groups have subgroups of orders of all of their divisors. The product of these subgroups of the factors will be a subgroup of G . $\square$

← converse of Lagrange's Thm is the case of abelian groups

Thm: If m is a **square free** integer, then every abelian group of order m is cyclic.

↑ not divisible by p^2 for any prime p

Pf:

FTFGAG: $G \simeq \mathbb{Z}_{p_1} \times \cdots \times \mathbb{Z}_{p_k}$ where the p_i are distinct primes.

These p_i are all relatively prime to each other, so

$G \simeq \mathbb{Z}_{p_1 \cdots p_k}$ is cyclic. $\square$